

האוניברסיטה העברית בירושלים
THE HEBREW UNIVERSITY OF JERUSALEM

WHEN IS UNSUPERVISED DISENTANGLEMENT POSSIBLE?

Daniella Horan, Eitan Richardson & Yair Weiss

NeurIPS 2021

OUTLINE

Motivation
- Disentanglement

Prior work
- Impossibility

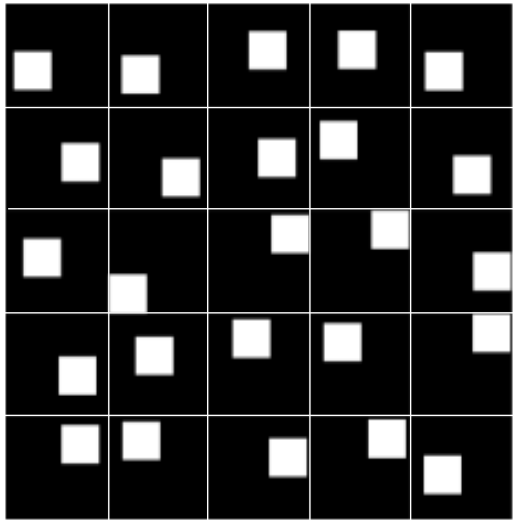
Our work
- Possibility theorem
- Experiments

Conclusion

MOTIVATION

$$z = \begin{bmatrix} horizontal\ position \\ vertical\ position \end{bmatrix}$$

g



$$z = \begin{bmatrix} eye\ color \\ hair\ color \\ nose\ size \\ smile \\ \vdots \end{bmatrix}$$

g

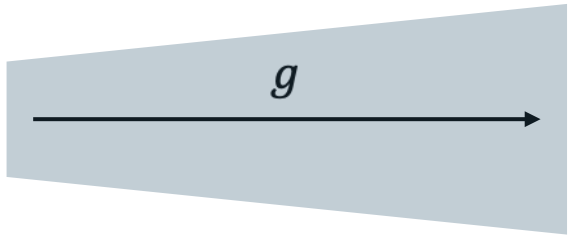


MOTIVATION

The (Unsupervised) **Manifold Hypothesis**:
"Real world data presented in high-dimensional spaces is likely to concentrate in the vicinity of non-linear sub-manifolds of much lower dimensionality"

Rifai, S., Dauphin, Y.N., Vincent, P., Bengio, Y. and Muller, X., 2011. The manifold tangent classifier.

$$z = \begin{bmatrix} eye\ color \\ hair\ color \\ nose\ size \\ smile \\ \vdots \end{bmatrix}$$





REPRESENTATION LEARNING

$$\begin{bmatrix} fur\ color \\ pose \end{bmatrix}$$

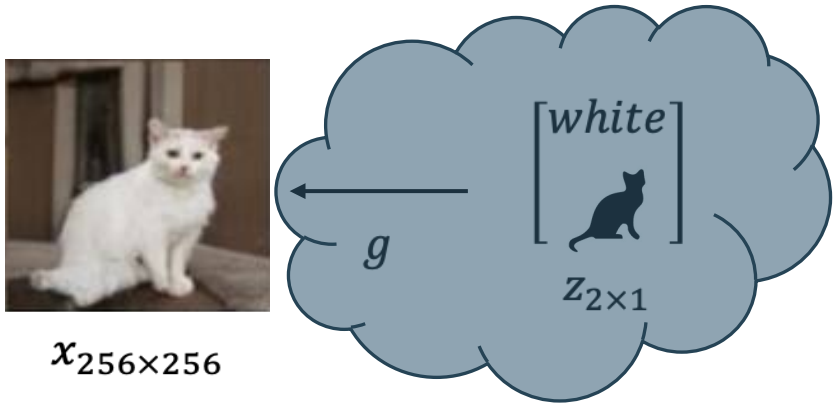
g



REPRESENTATION LEARNING



Input

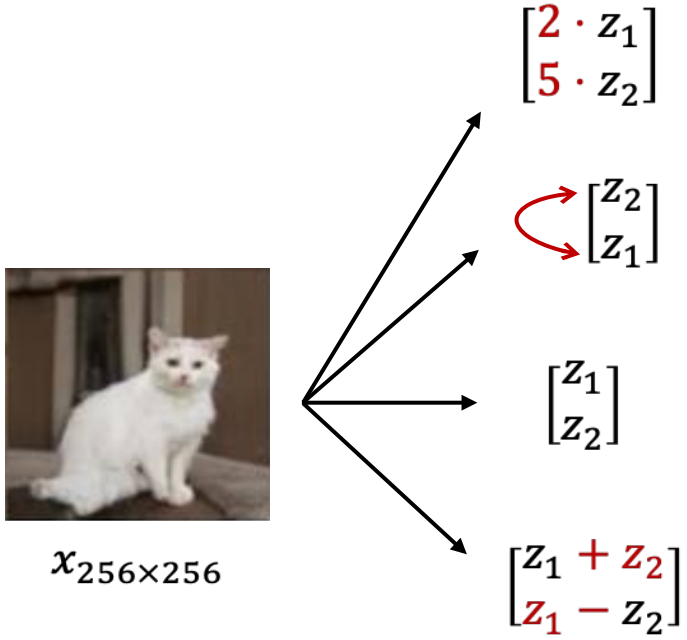


Learned
Encoder

REPRESENTATION LEARNING



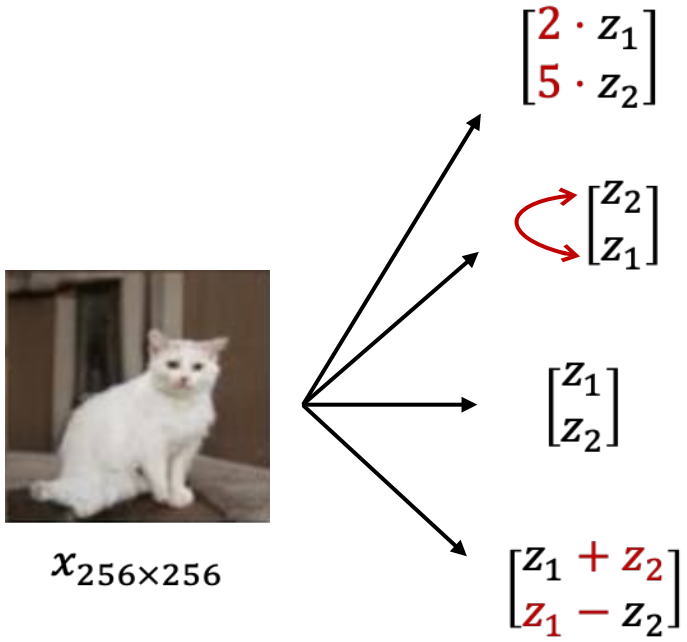
Input



Learned
Encoder

DISENTANGLEMENT

- Separation of the factors of variations in the data
- Change in single true factor = change in single learned factor

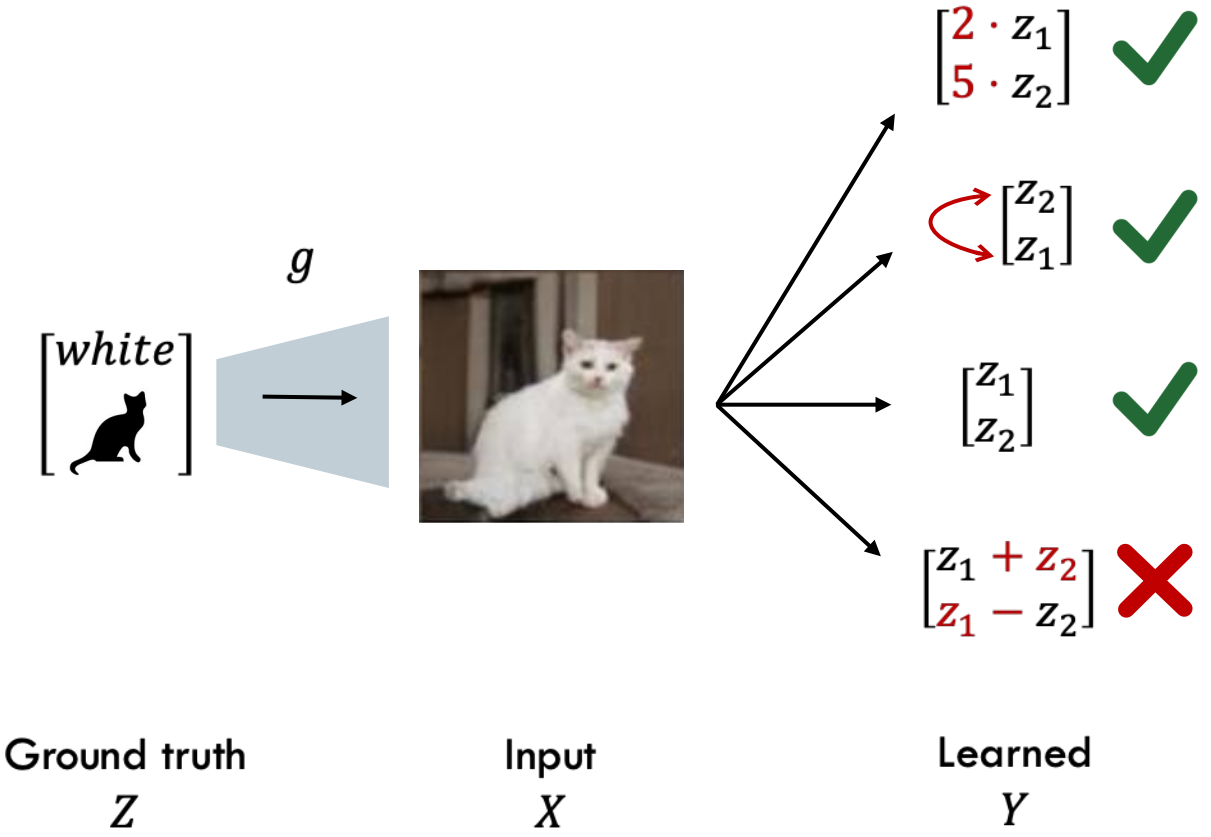


DISENTANGLEMENT

Let $\boldsymbol{x} \in \mathbb{R}^{\boldsymbol{n}}$, $\boldsymbol{z} \in \mathbb{R}^{\boldsymbol{d}}$, $\boldsymbol{g}: \mathbb{R}^{\boldsymbol{d}} \to \mathbb{R}^{\boldsymbol{n}}$ s.t. $\boldsymbol{x} = \boldsymbol{g}(\boldsymbol{z})$.

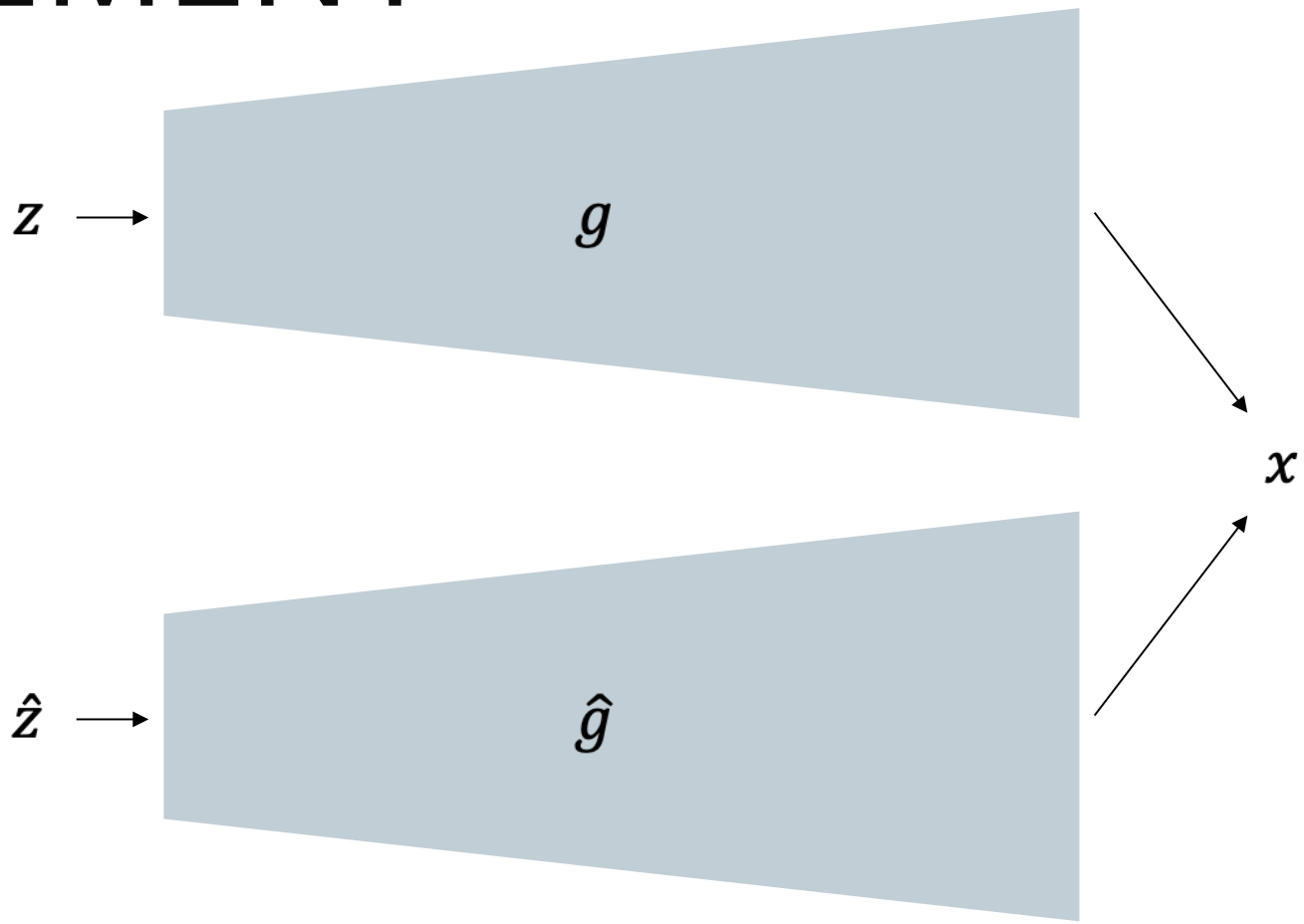
Then $\boldsymbol{y} \in \mathbb{R}^{\boldsymbol{d}}$ is a ***disentangled representation*** of z if there exist a pointwise function $f : \mathbb{R}^{d} \to \mathbb{R}^{\mathrm{d}}$, and permutation $\pi: \{d\} \to \{d\}$ s.t.

$$Y = \pi(f(Z))$$



IS UNSUPERVISED DISENTANGLEMENT POSSIBLE?

Assumptions:

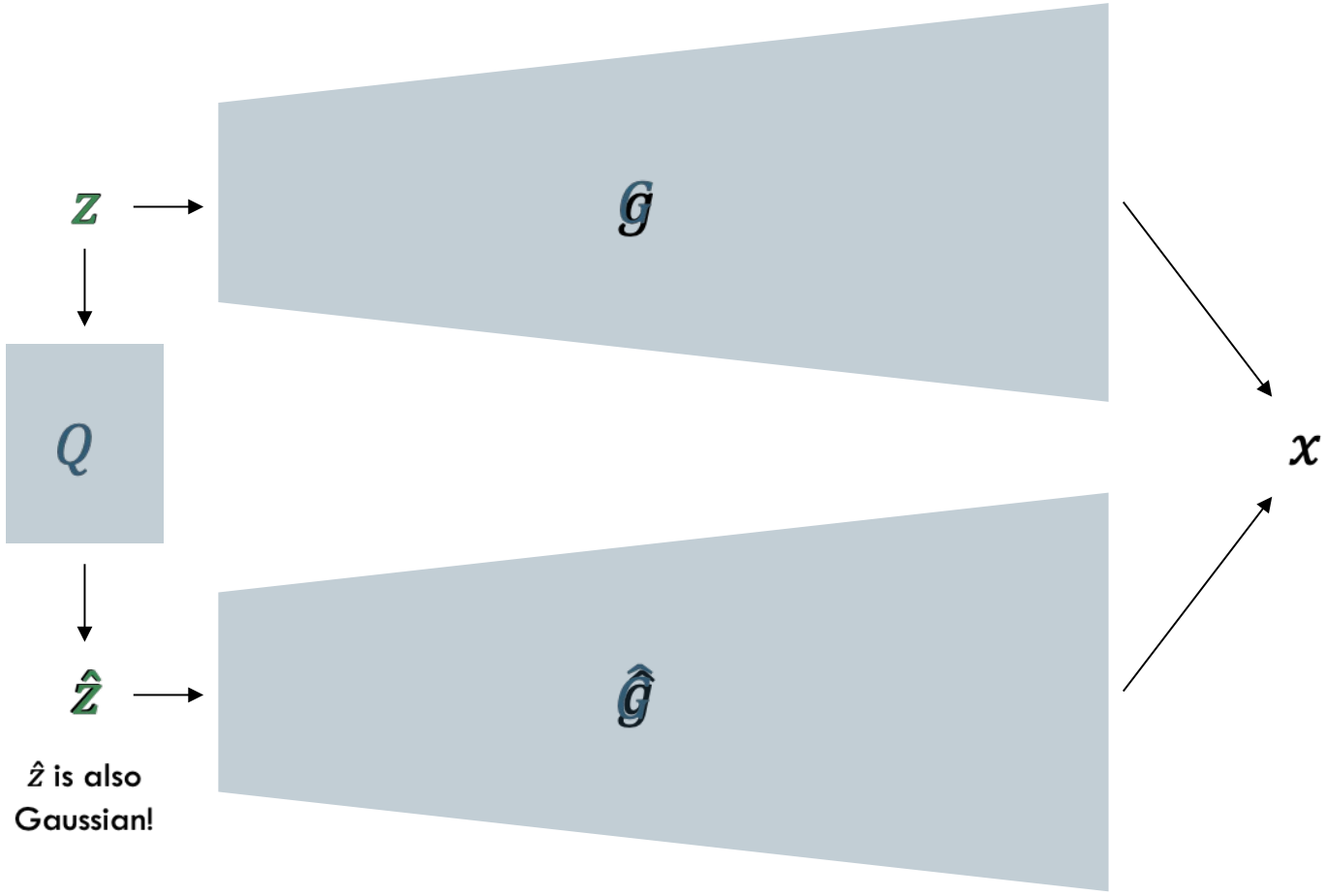


Hyvärinen, A. and Oja, E., 2000. Independent component analysis: algorithms and applications. *Neural networks*, *13*(4-5), pp.411-430.

IS IT POSSIBLE: THE LINEAR CASE

Assumptions:

1. The generator is linear
2. Gaussian prior on the latent distribution

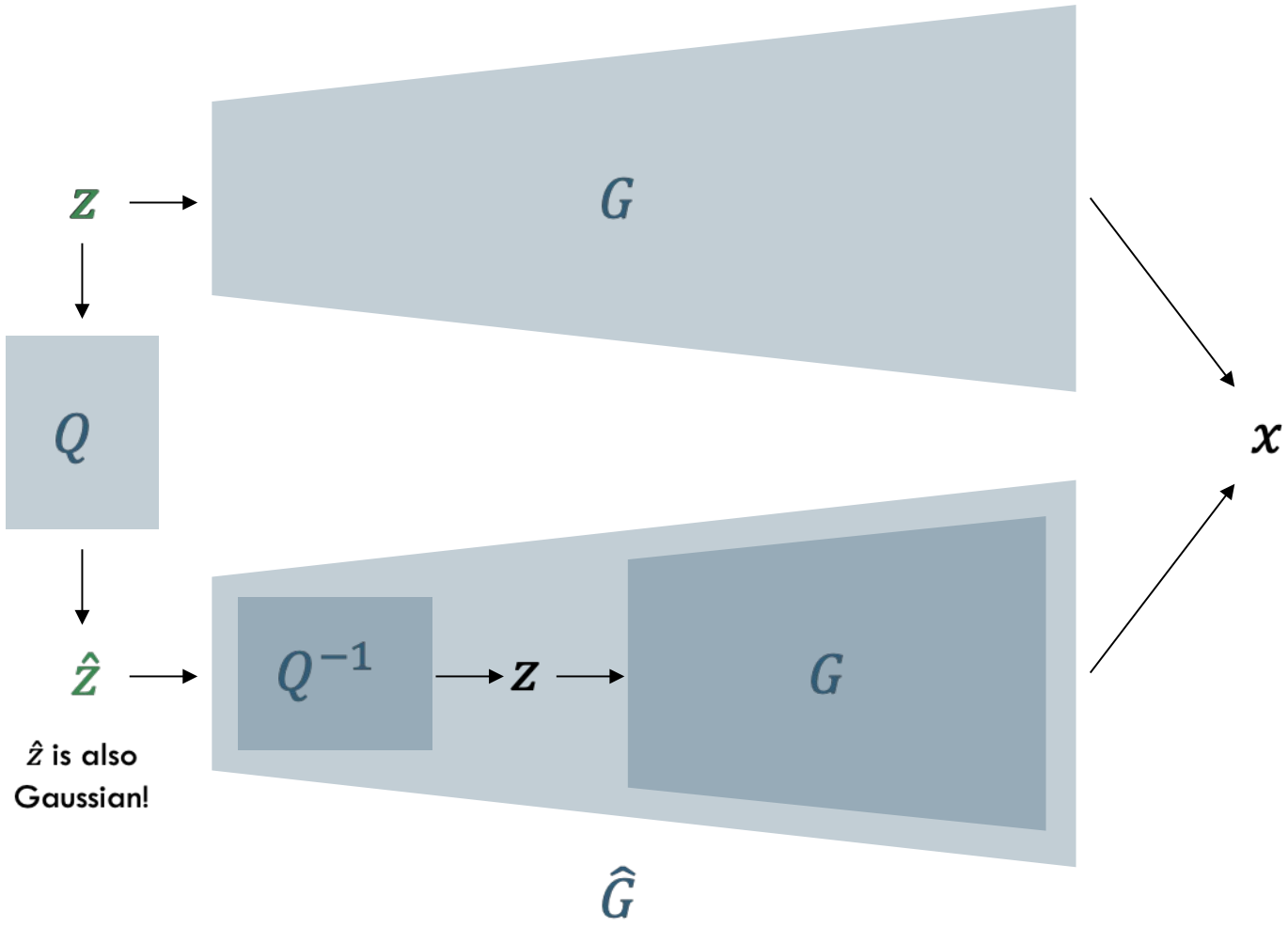


Hyvärinen, A. and Oja, E., 2000. Independent component analysis: algorithms and applications. *Neural networks*, *13*(4-5), pp.411-430.

IS IT POSSIBLE: THE LINEAR CASE

Assumptions:

1. The generator is linear
2. Gaussian prior on the latent distribution



Hyvärinen, A. and Oja, E., 2000. Independent component analysis: algorithms and applications. *Neural networks*, *13*(4-5), pp.411-430.

IS IT POSSIBLE: THE LINEAR CASE

Assu

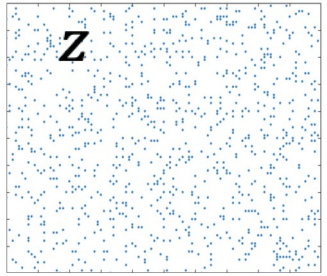
1. .

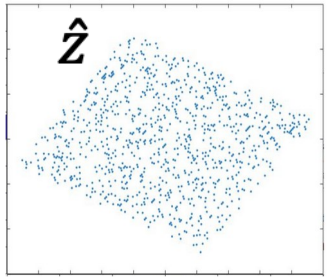
non-

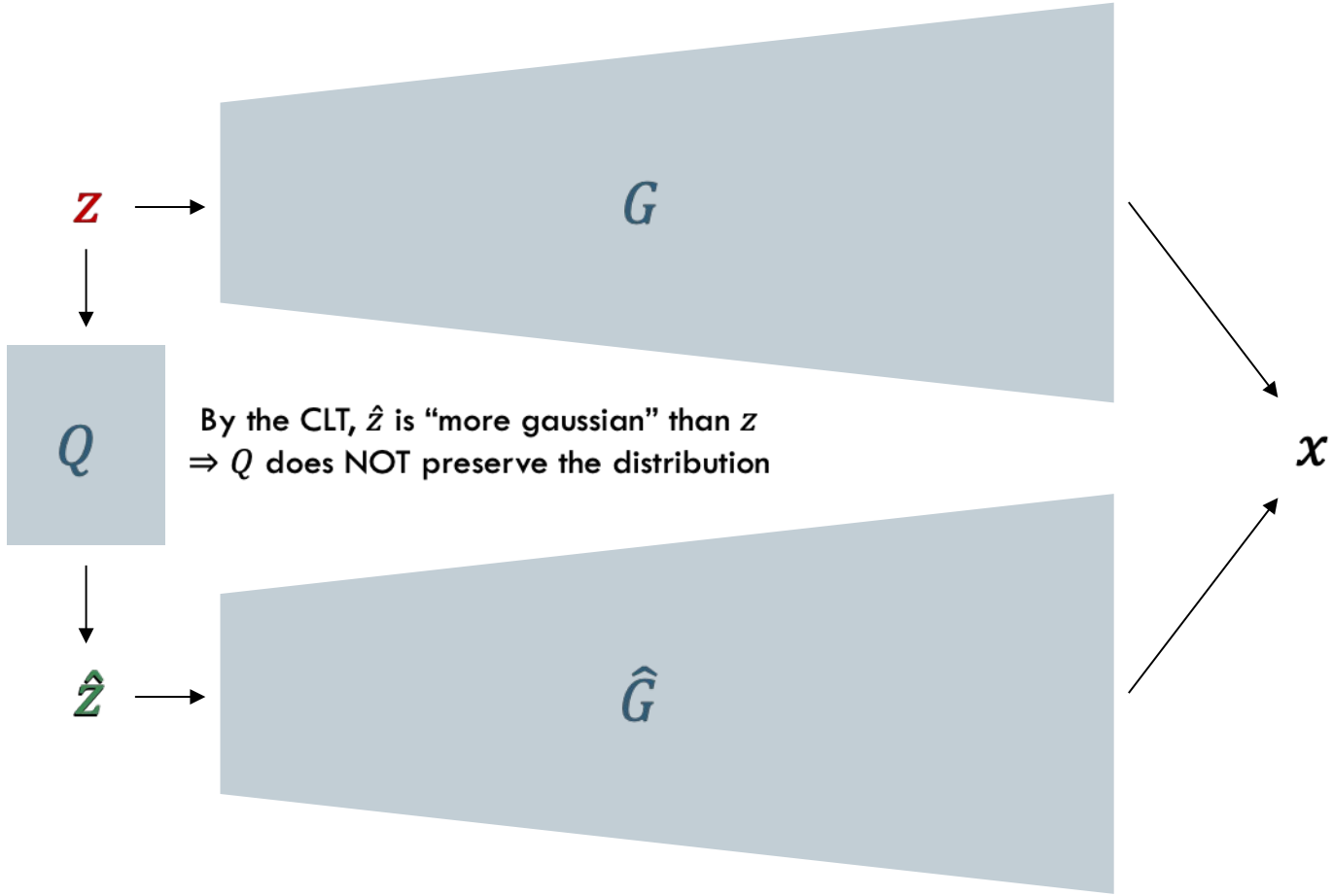
2. s

on

the latent distribution







Hyvärinen, A. and Oja, E., 2000. Independent component analysis: algorithms and applications. *Neural networks*, *13*(4-5), pp.411-430.

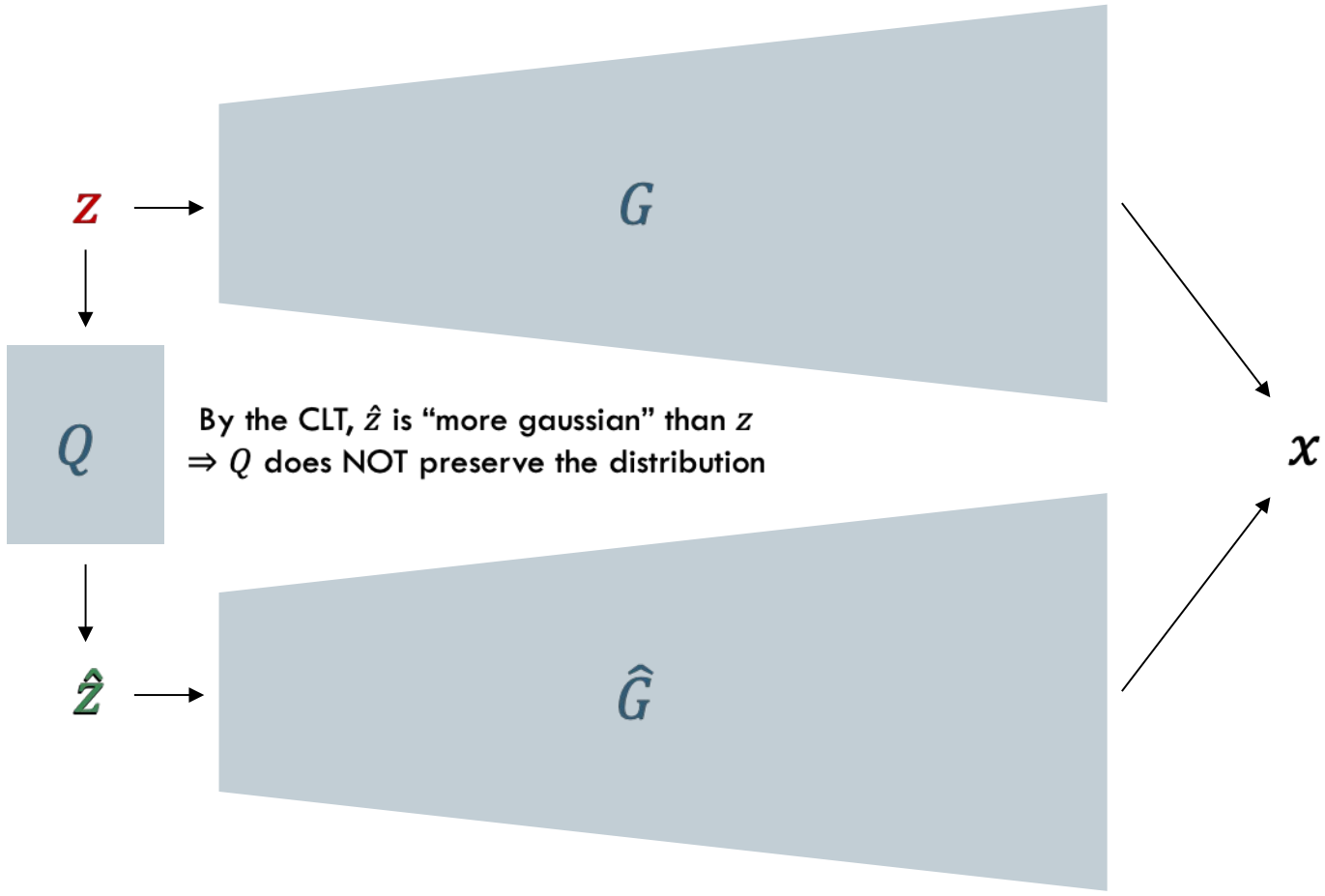
IS IT POSSIBLE: THE LINEAR CASE

Assumptions:

1. The generator is non-linear
2. Gaussian prior on the latent distribution

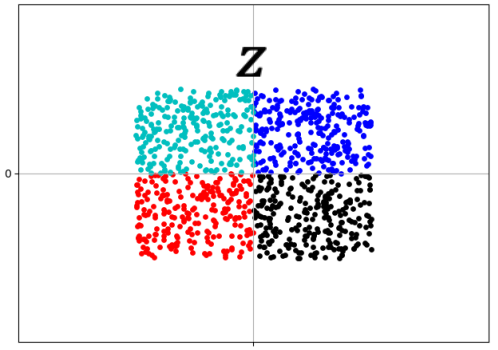
⇒ Under these assumptions, the answer is YES!
Disentanglement *IS* possible

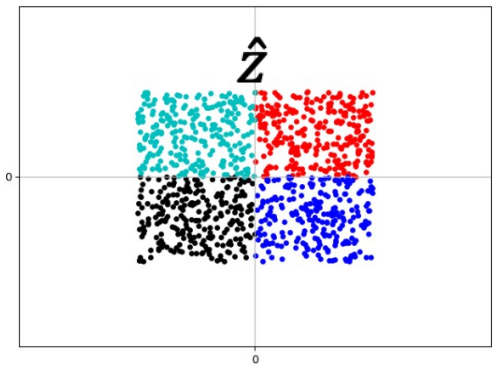
But they are very strong…

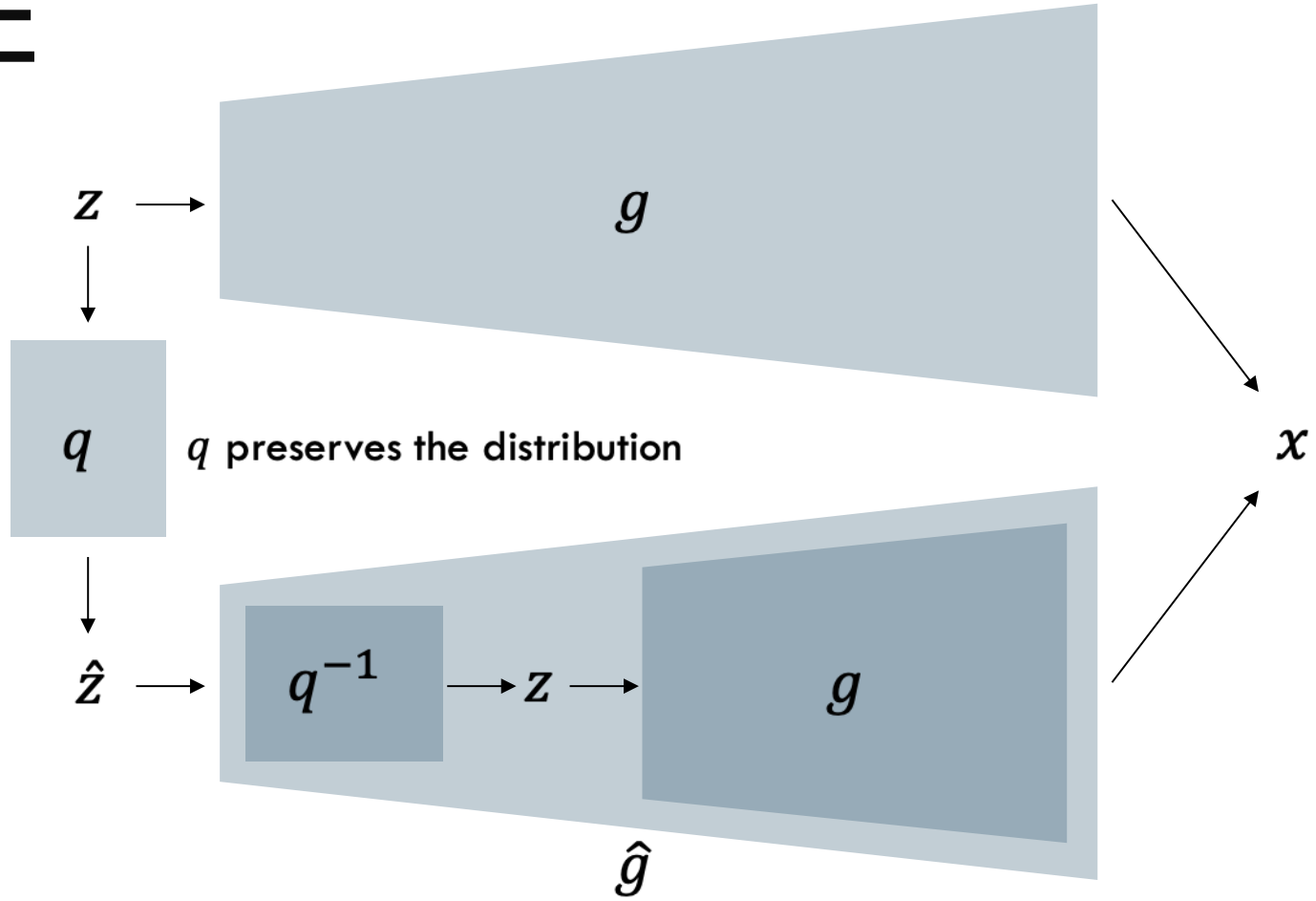


Hyvärinen, A. and Oja, E., 2000. Independent component analysis: algorithms and applications. *Neural networks*, *13*(4-5), pp.411-430.

IS IT POSSIBLE: THE NON-LINEAR CASE







In the general case, the answer is NO!

Locatello, F., Bauer, S., Lucic, M., Raetsch, G., Gelly, S., Schölkopf, B. and Bachem, O., 2019, May. Challenging common assumptions in the unsupervised learning of disentangled representations.

IS IT POSSIBLE: THE NON-LINEAR CASE

"unsupervised learning of disentangled representations is fundamentally impossible without inductive biases on both the models and the data"

Locatello, F., Bauer, S., Lucic, M., Raetsch, G., Gelly, S., Schölkopf, B. and Bachem, O., 2019, May. Challenging common assumptions in the unsupervised learning of disentangled representations.

In the general case, the answer is NO!

IS DISENTANGLEMENT POSSIBLE?

Recap:

In the linear case: YES

In the non-linear case: Without further assumptions, NO

Our question:

Are there assumptions that can guarantee possibility in the general case?

If so, are they reasonable?

SHORT ANSWER: YES

We prove that under the assumptions of:

- *Local Isometry* (on the model)
- *non-Gaussianity* (on the latent distribution)

Unsupervised disentanglement is possible,

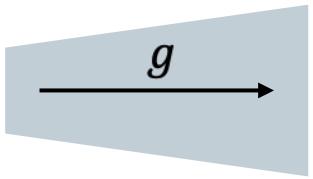
using the method of HLLE followed by ICA.

LOCAL ISOMETRY

Intuitively: "ϵ change in $Z \cong \epsilon$ change in X"

$$\begin{bmatrix} z_1 = whi \\ z_2 = \text{  } \end{bmatrix}$$

$z_{2\times1}$





$x_{256\times256}$

Formally: $g: Z \to X$ is *locally isometric* if for all z, the Jacobian $J = \frac{\partial g}{\partial z}$ satisfies:

$$J(z)^T J(z) = I$$

{ If linear: $J = G$

Locally linear → ϵ change }

THE METHOD

Step 1: Hessian Eigenmaps (HLLE)

- Find embedding that minimizes the curvature (≅ Hessian)
- Solve eigenproblem

Step 2: Fast Independent Component Analysis (FastICA)

- Find linear representation that minimizes Gaussianity
- Iterative process

Donoho, D.L. and Grimes, C., 2003. Hessian eigenmaps: Locally linear embedding techniques for high-dimensional data.
Hyvarinen, A., 1999. Fast and robust fixed-point algorithms for independent component analysis.

POSSIBILITY THEORFM

Let $\{x_i\}$ be samples from a manifold $x = g(z)$. If:

1. There exists an alternative representation of the manifold $\mathrm{x} = g_2(s)$ with $s = Af(z)$, for any matrix A and pointwise nonlinearity f, such that g_2 satisfies ***local isometry***
2. $f(z)$ is sufficiently ***non-Gaussian***

Then as the number of samples goes to infinity, HLLE+ICA will **provably find a disentangled representation, *y*, of the true latent factors *z***.

PROOF OUTLINE

HLLE

- g_2 satisfies local isometry
- Infinite data

⇒ Guaranteed to find s up to rigid motion

FastICA

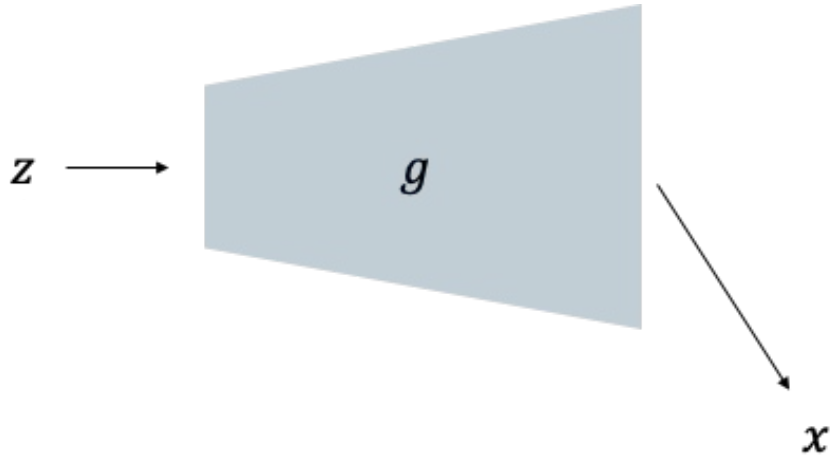
- $y = f(z)$ is sufficiently non-Gaussian
- A is a linear transformation

⇒ Guaranteed to find y

Disentanglement

- f is a pointwise function

⇒ y is a disentangled representation of z



Donoho, D.L. and Grimes, C., 2003. Hessian eigenmaps: Locally linear embedding techniques for high-dimensional data.
Hyvarinen, A., 1999. Fast and robust fixed-point algorithms for independent component analysis.

IS DISENTANGLEMENT POSSIBLE?

Recap:

In the linear case: YES

In the non-linear case: Without further assumptions, NO

Our question:

Are there assumptions that can guarantee possibility in the general case? ✔

If so, are they reasonable?

IS LOCAL ISOMETRY REASONABLE?

- Any manifold can be well approximated using an isometric parameterization (Nash-Kuiper, 1956)
- Provably holds for commonly used manifolds (e.g. faces, shapes) (Donoho-Grimes, 2003)
- Widely used in Manifold Learning (e.g. LLE, HLLE, Isomap)

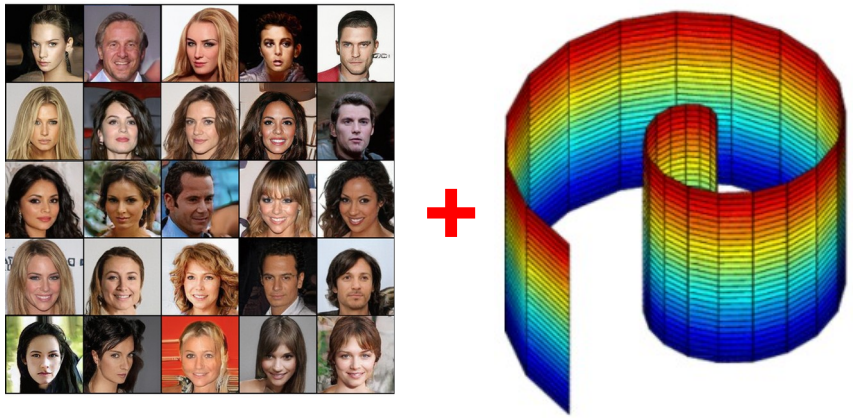
Nash, J., 1956. The imbedding problem for Riemannian manifolds.
Roweis, S.T. and Saul, L.K., 2000. Nonlinear dimensionality reduction by locally linear embedding.
Donoho, D.L. and Grimes, C., 2003. Hessian eigenmaps: Locally linear embedding techniques for high-dimensional data.
Tenenbaum, J.B., De Silva, V. and Langford, J.C., 2000. A global geometric framework for nonlinear dimensionality reduction.

EXPERIMENTS

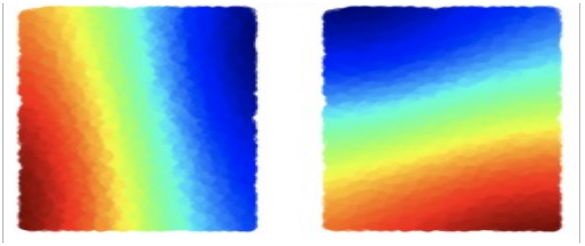
How do we measure disentanglement?

Realistic

Manifolds with ground truth



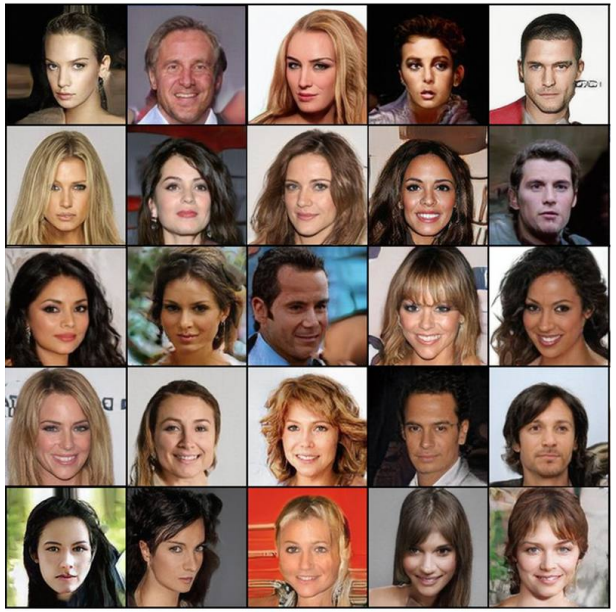
Disentanglement visualization & evaluation



MCC

REALISTIC MANIFOLD GENERATING SCHEME

$$z = \begin{bmatrix} eye\ color \\ hair\ color \\ nose\ size \\ smile \\ \vdots \end{bmatrix} \xrightarrow{\quad g \quad}$$



Härkönen, E., Hertzmann, A., Lehtinen, J. and Paris, S., 2020. Ganspace: Discovering interpretable gan controls.
Karras, T., Laine, S., Aittala, M., Hellsten, J., Lehtinen, J. and Aila, T., 2020. Analyzing and improving the image quality of stylegan.

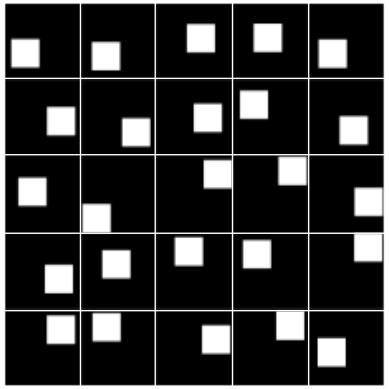
REALISTIC MANIFOLD GENERATING SCHEME

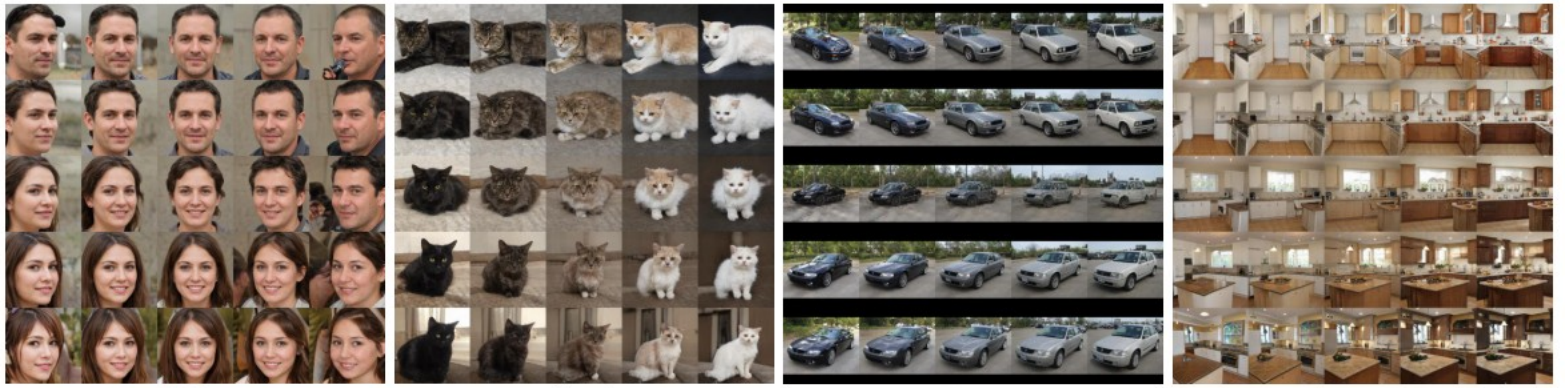
$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \xrightarrow{PCA} v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_k \end{bmatrix} \xrightarrow{fix\ g} z = \begin{bmatrix} v_6 \approx pose \\ v_9 \approx color \end{bmatrix} \xrightarrow{g}$$

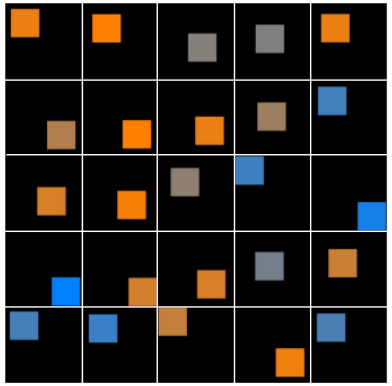


Härkönen, E., Hertzmann, A., Lehtinen, J. and Paris, S., 2020. Ganspace: Discovering interpretable gan controls.
Karras, T., Laine, S., Aittala, M., Hellsten, J., Lehtinen, J. and Aila, T., 2020. Analyzing and improving the image quality of stylegan.

MANIFOLDS

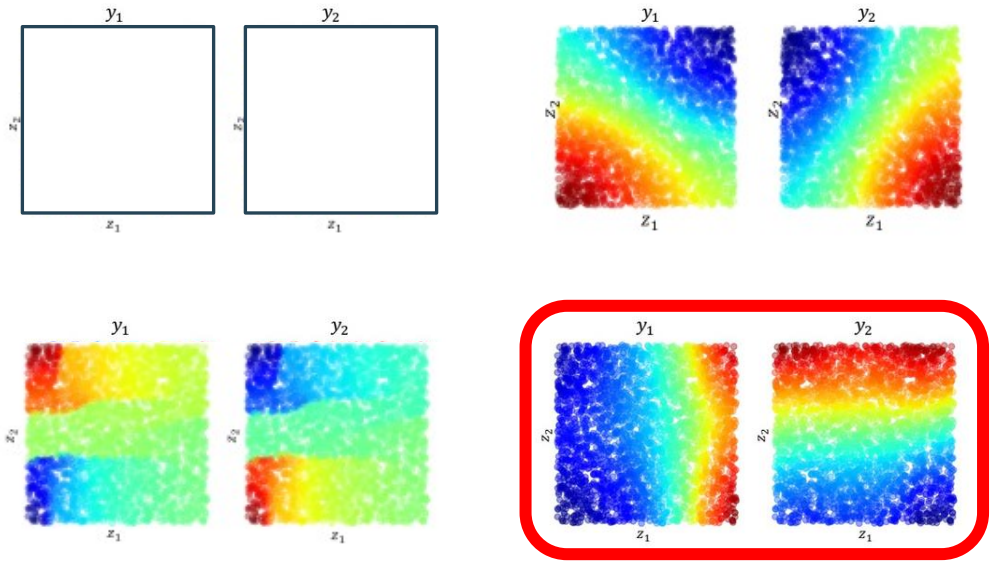


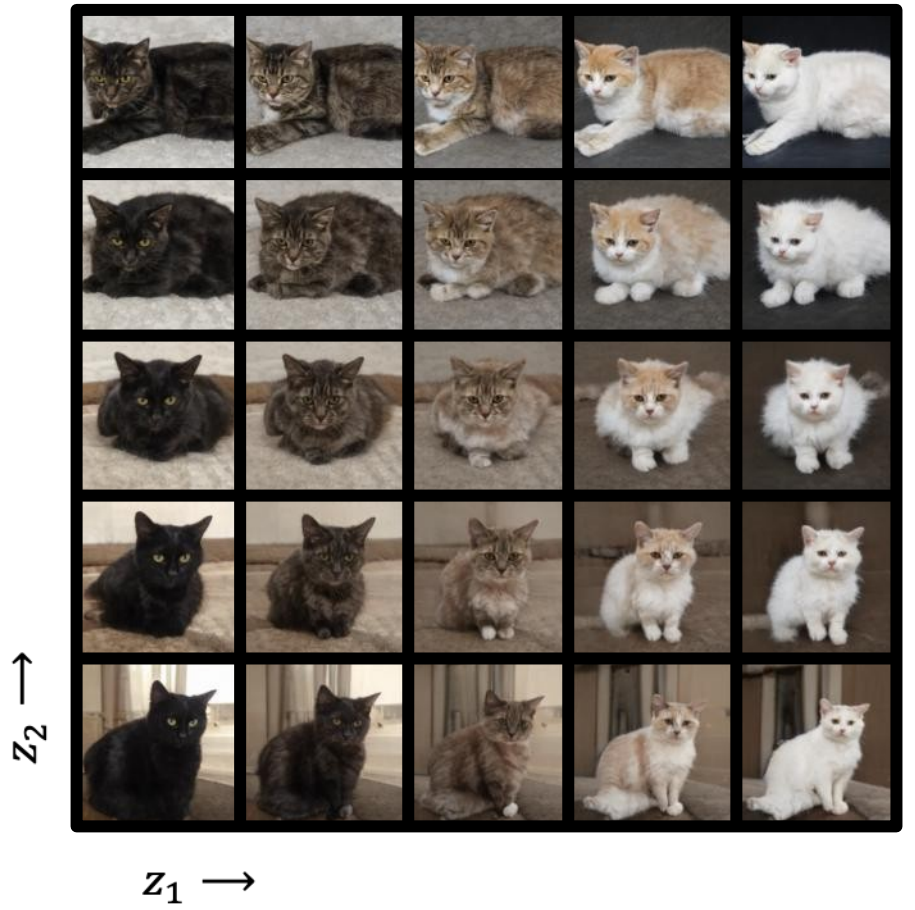






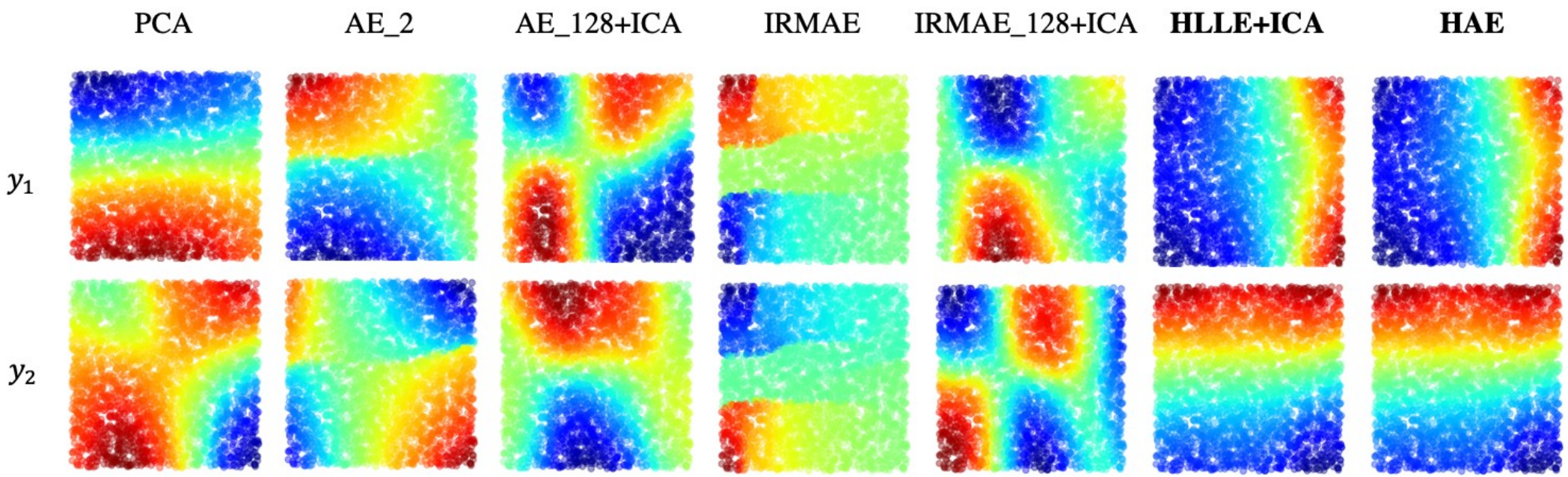
DISENTANGLEMENT EVALUATION





QUALITATIVE RESUL



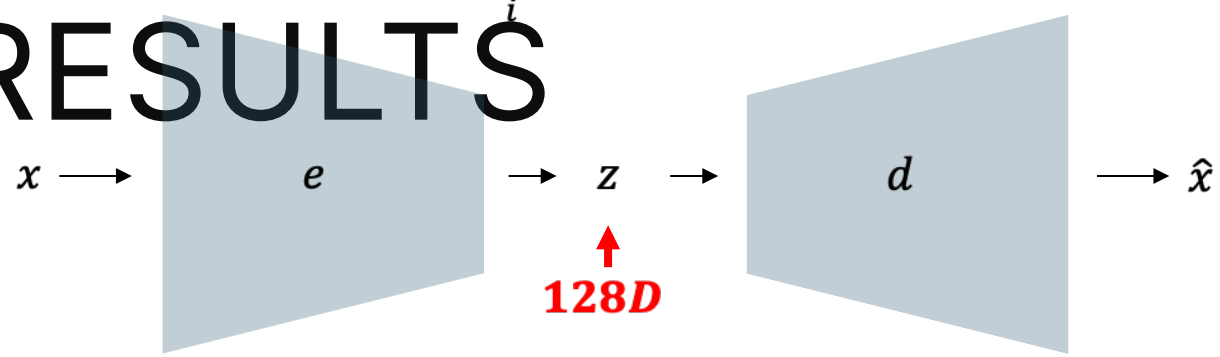


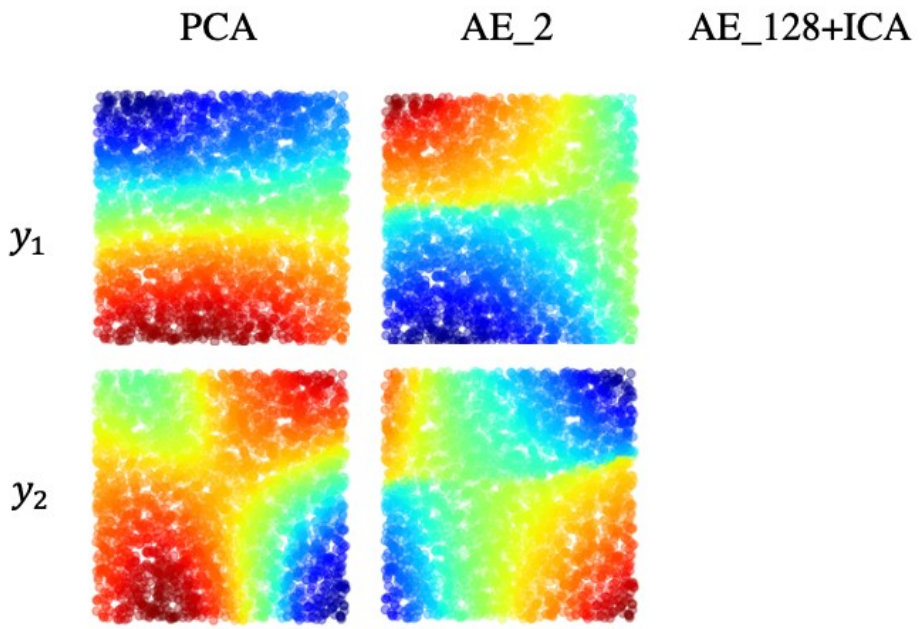
QUALITATIVE RESULTS



QUALITATIVE RESULTS

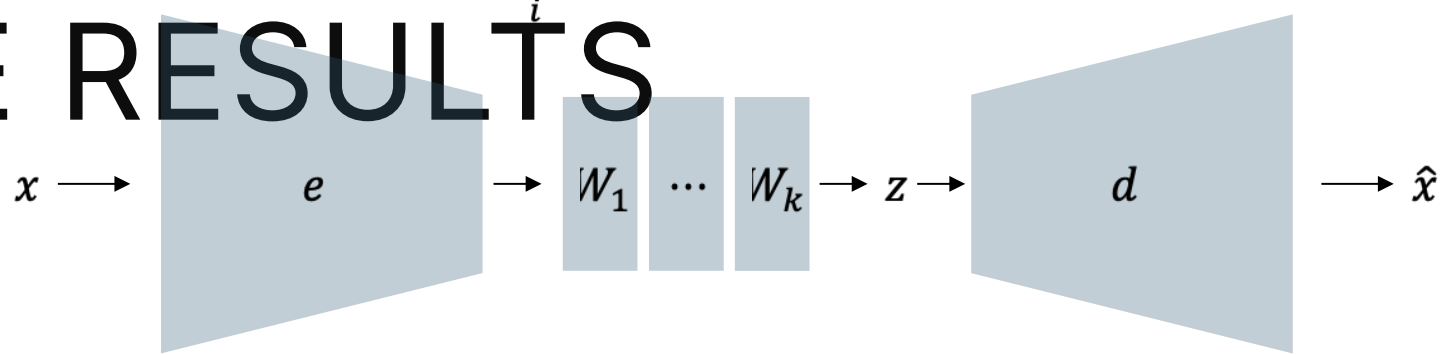
$$L(x) = \sum_i \left\| \mathcal{D}\left(\mathcal{E}(x_i)\right) - x_i \right\|^2$$

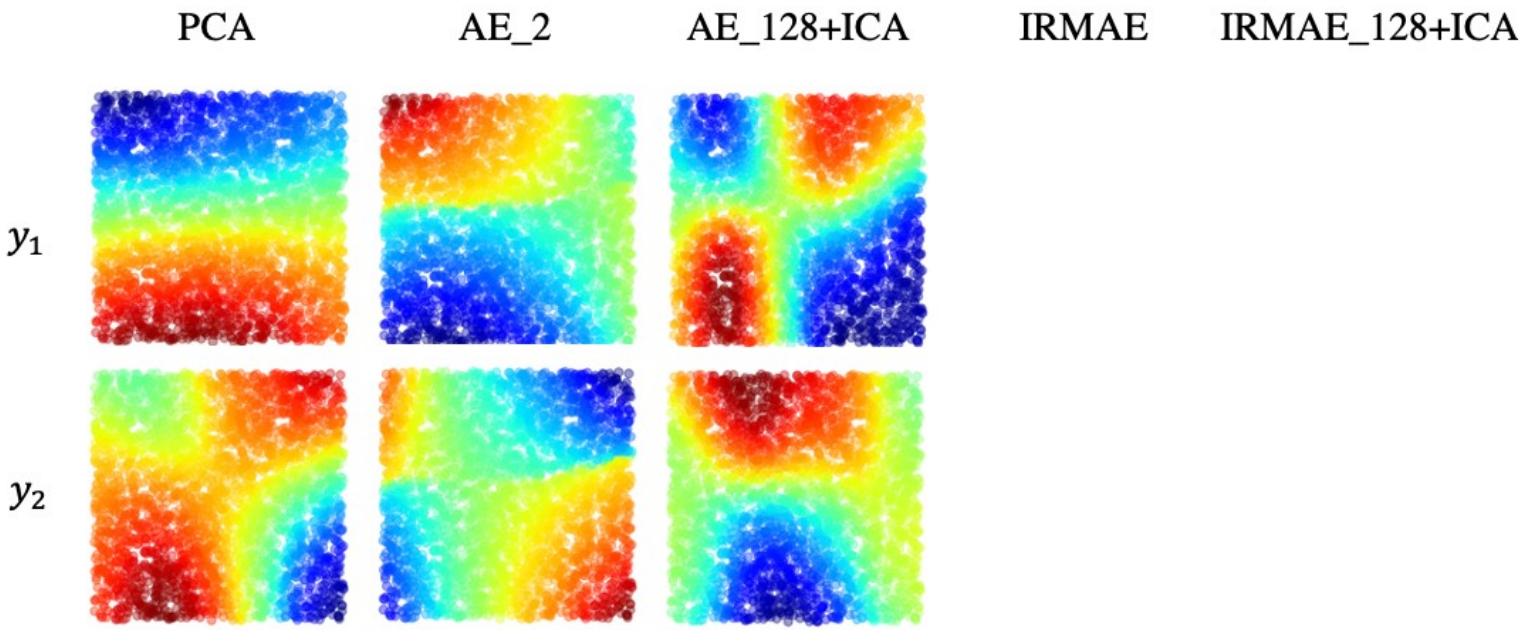




QUALITATIVE RESULTS

$$L(x) = \sum_i \left\| \mathcal{D}\left(W_k \dots W_1 \mathcal{E}(x_i)\right) - x_i \right\|^2$$





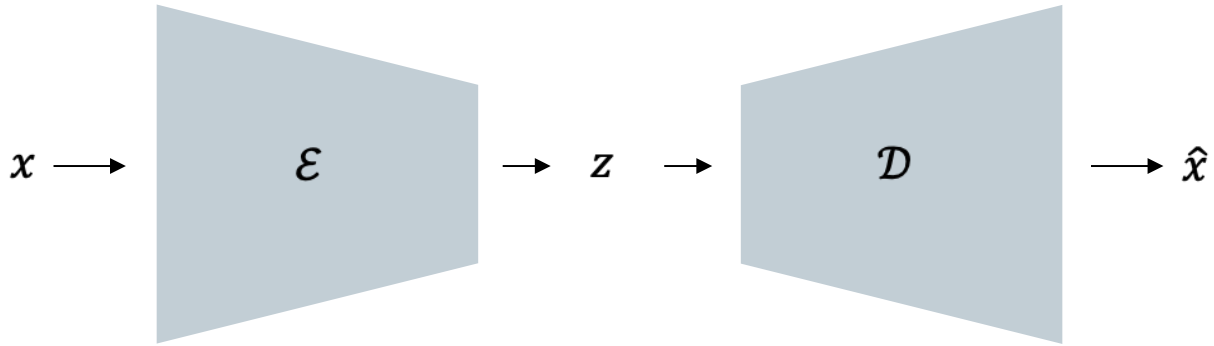
Jing, L., Zbontar, J. and LeCun, Y., 2020. Implicit rank-minimizing autoencoder.

HESSIAN AUTO-ENCODER (HAE)

HLLE_ICA	AE
✅	❌

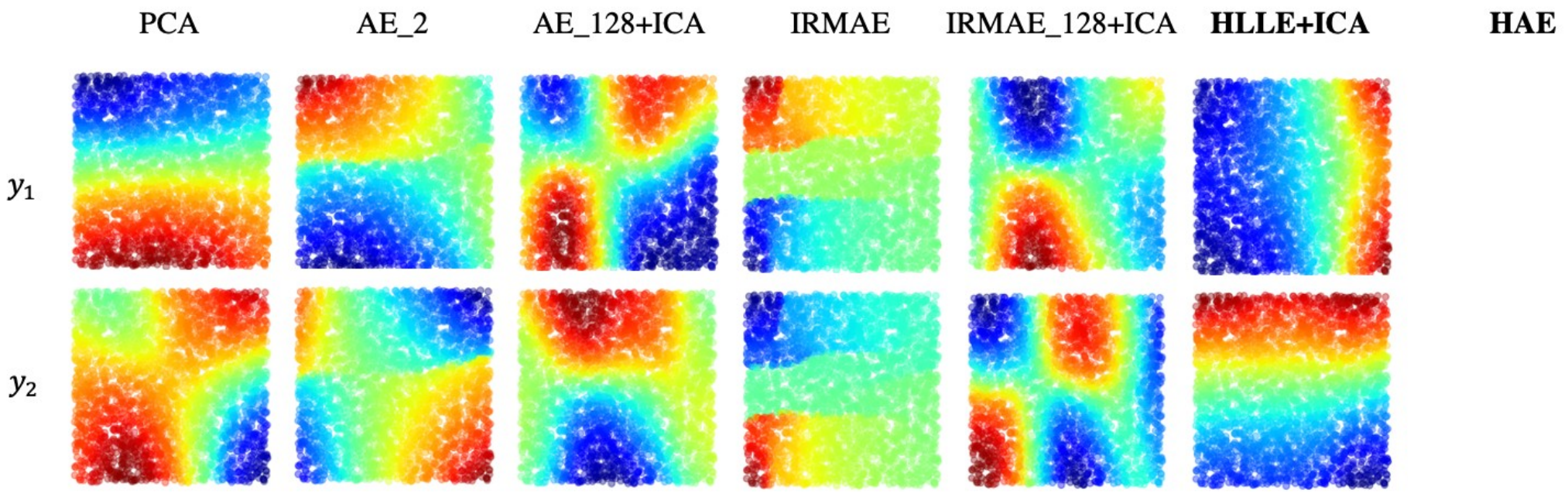
Student-teacher paradigm:

$$Y = HLLE_ICA(X)$$



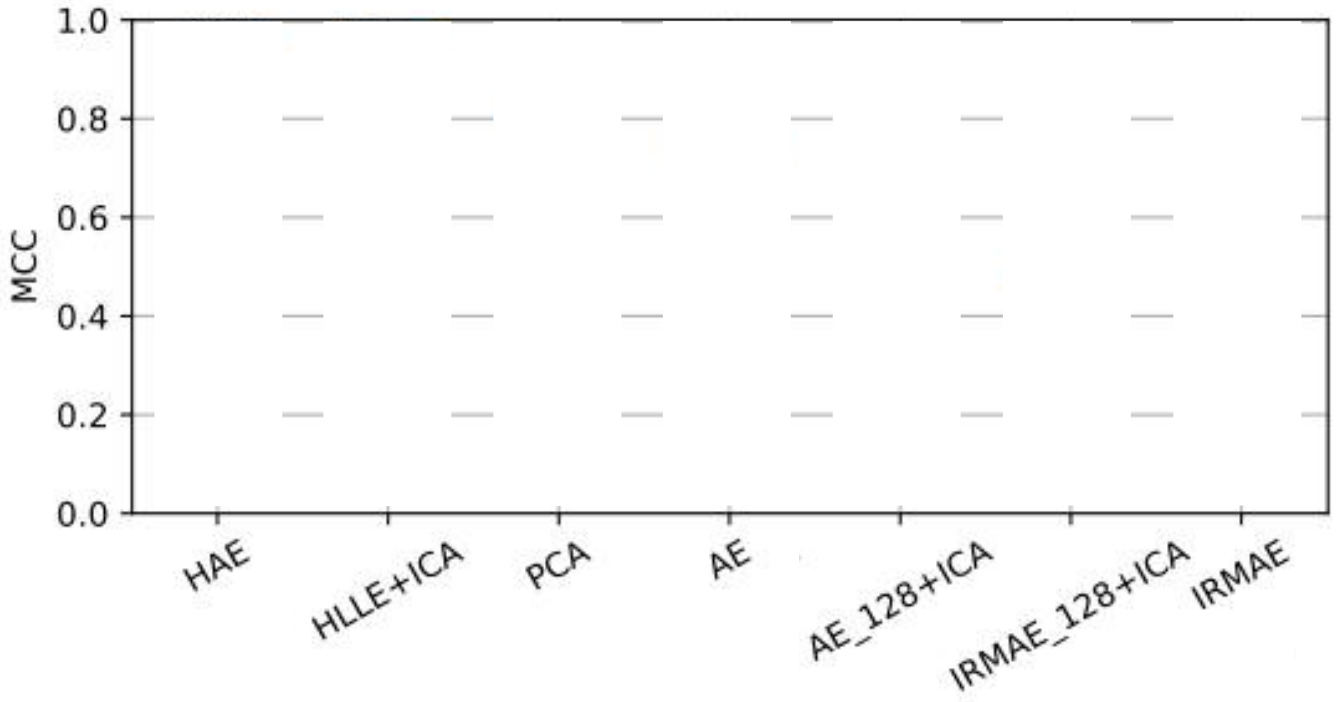
$$L(x, y) = \sum_i \|\mathcal{E}(x_i) - y_i\|^2 + \|\mathcal{D}(y_i) - x_i\|^2$$

QUALITATIVE RESULTS

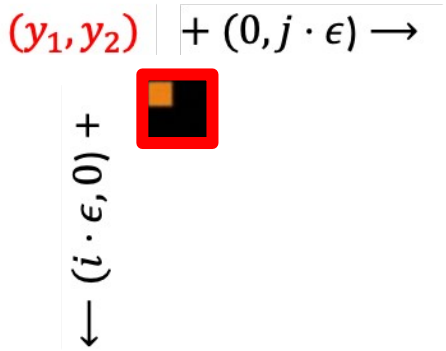


QUANTITATIVE RESULTS

Mean Correlation Coefficient (MCC)

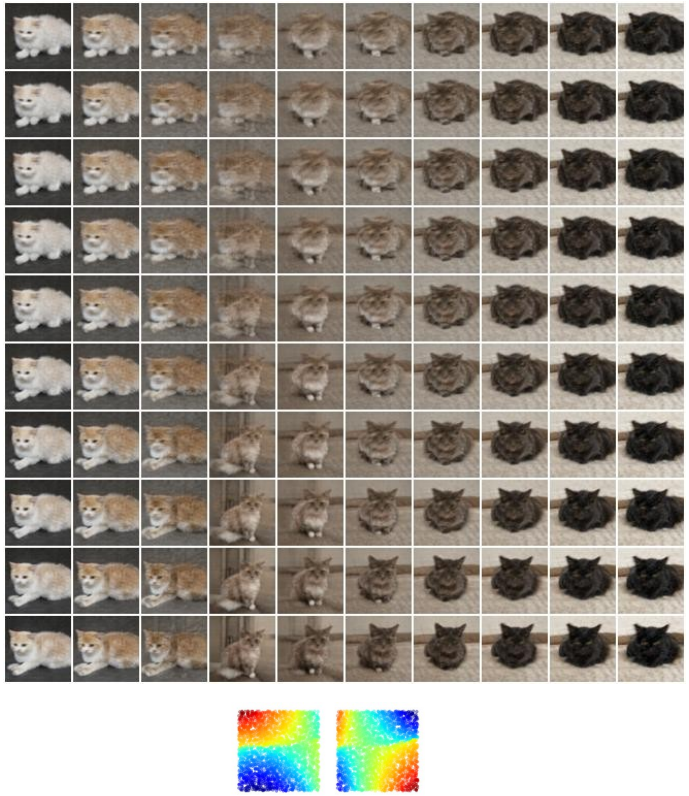


INTERPOLATIONS

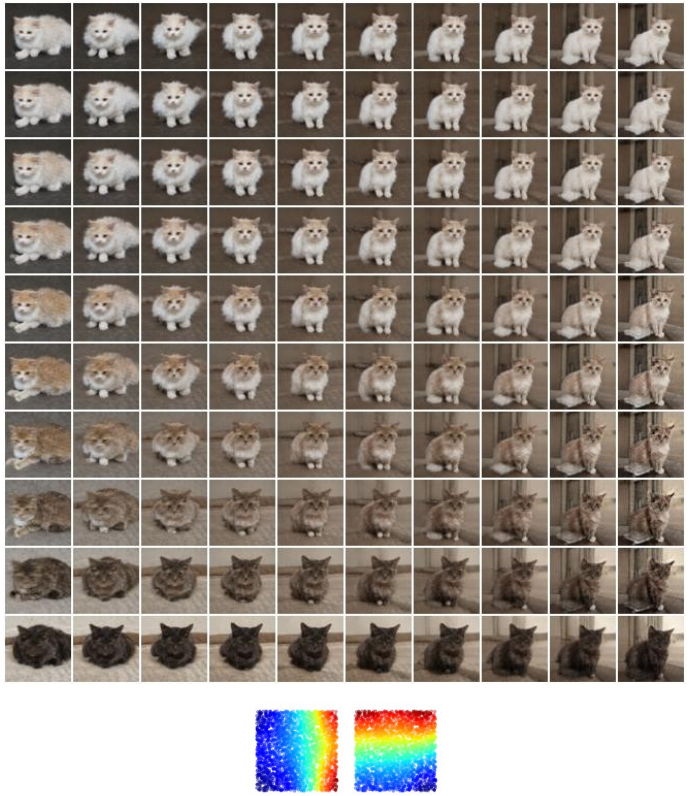


INTERPOLATIONS

AE



HAE



LIMITATIONS

Infinite data (\dense sampling)

- Sparse sampling?

Spectral method

- Large datasets?

CONCLUSIONS

Possibility Theorem

- Under local isometry and non-Gaussianity, unsupervised disentanglement is provably possible

Experiments

- Realistic manifolds seem to satisfy the assumptions
- Methods that assume local isometry (HLLE_ICA, HAE) succeed in disentanglement, while baseline autoencoders fail

THANKS FOR LISTENING!